

Low cost GAN depth estimation for 3D object modeling

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Abstract—Depth estimation, one of the leading research in computer vision, from autonomous driving to 3D modeling, play key role to artificial perception of the machine. This paper present reconstruction of 3D object with the optimal cost for accuracy, using simple Generative Adversarial Network (GAN) on NYU Depth v2 and KITTI dataset. Our approach in optimization achieved xx% accuracy with only xx epochs training resulted in xx times yield. We concluded this approach cost very inexpensive compared to other method.

Index Terms—Depth Estimation, Generative Adversarial Network, 3D Modeling

I. Introduction

Depth estimation (DE) research in robotic [1], autonomous driving, and 3d modeling has varied in result with overall accuracy consistently high. There are many approach explored by the current researches from optimizing hardware [1], improving dataset [2], and/or using new set of architecture [2, 3]. The common approach on the current research is using deep learning (DL) as primary architecture in research.

DL is still the leading architecture model for monocular depth estimation (MDE) [4]. There are many variety used on MDE such as convolutional neural networks (CNNs), recurrent neural net-works (RNNs), variational auto-encoders (VAEs) and generative adversarial networks (GANs) [5]. Eigen et al. [6] are the pioneers for using CNN for MDE, by using local and global information from single image. GAN has become a hot research direction in recent years [5], commonly use generator to estimate global and local 3D structure, and discriminator to predict depth of monocular image. GAN also varied in framework, such as stacked GAN, conditional GAN [3] and Cycle GAN [5].

These varied method and framework still face a critical problem, high economic cost to create data-sets with ground truth [5], and high cost of tra-ining. Our approach aims to create 3D object modeling and reduce the cost of training, optimizing cost to accuracy and error function using GAN architecture framework, leveraging on GAN ability for capturing global structure with fewer training data than a standard en-coder-decoder CNN [3].

II. Related Work

Related work goes here.

III. Proposed Method

Proposed method on research.

IV. Experiment

Experiment details.

V. Discussion

Discussion about the results, and ablation study.

VI. Conclusion

Conclusion goes here.

VII. Acknowledgments

Aknowledgments goes here.

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